

6.1 总体与样本

总体 $\rightarrow$ 全体

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有限总体 无限总体

* 总体的变量为 X , 则 X 的分布为
总体分布.

样本: 抽样: 总体中抽一部分

样本容量 (n): 取几个

抽 n 次 $(X_1, X_2, \dots, X_n)$ 变量.

抽出 n 个 $(X_1, X_2, \dots, X_n)$ 观测值
(确定)

简单随机抽样

- ①: 与 X 有相同的分布, 每个个体被抽到的机会是均等的
- ②: 相互独立

样本的分布 $(X_1, X_2, \dots, X_n)$

联合分布函数: $F(x_1, x_2, \dots, x_n) = F(x_1)F(x_2)\dots F(x_n)$

若为离散型 $P(X_1=x_1, \dots, X_n=x_n) = P(X_1=x_1)\dots P(X_n=x_n)$

若为连续型 $f(x_1, x_2, \dots, x_n) = f(x_1)\dots f(x_n)$

例: X : 0-1 分布. 参数为 p . $(X_1, \dots, X_n)$ 是样本

$$P(X=x) = p^x (1-p)^{1-x} \quad x=0, 1$$

$$P(X_1=x_1, X_2=x_2, \dots, X_n=x_n) = P(X_1=x_1)\dots P(X_n=x_n)$$

$$= p^{x_1} (1-p)^{1-x_1} \cdot p^{x_2} (1-p)^{1-x_2} \dots p^{x_n} (1-p)^{1-x_n}$$

$$= p^{x_1+x_2+\dots+x_n} (1-p)^{n-(x_1+x_2+\dots+x_n)}$$

6.2.1 统计量的定义

定义: 不含任何未知参数的 样本 的函数
构造.

如:

$$X_1 + X_2 + \dots + X_n$$

$$X_1^2 + X_2^2 + \dots + X_n^2$$

$$\max\{|X_1|, |X_2|, \dots, |X_n|\}$$

$$X_1$$

$$\min\{X_3, X_4\}$$

不是统计量:

$$N(\mu, \sigma^2) \quad \mu, \sigma \text{ 未知}$$

$$\sum_{i=1}^n (x_i - \mu)^2 \rightarrow \text{不是! } \mu \text{ 未知}$$

$$\frac{1}{n\sigma^2} (x_1 + \dots + x_n) \rightarrow \text{不是!}$$

$$\text{注: } \frac{X_1 + X_2 + \dots + X_n}{n} \rightarrow \text{是!}$$

6.2.2 常用统计量

设样本 $(X_1, X_2, \dots, X_n)$ 来自总体 X .

有 $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ (样本均值)

未修正的样本方差: $S_0^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$

(修正的) 样本方差: $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

$$\text{样本标准差: } S = \sqrt{S^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\text{样本 } k \text{ 阶原点矩: } A_k = \frac{1}{n} \sum_{i=1}^n X_i^k$$

$$\downarrow \text{—— 中心矩: } B_k = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^k$$

- 阶原点矩即为均值 ($k=1$)

= 阶中心矩即为未修正的
样本方差

* 协方差 $S_{12} = \frac{1}{n} \sum (X_i - \bar{X})(Y_i - \bar{Y})$

$$R = \frac{S_{12}}{S_1 S_2}$$

例: 泊松分布. 参数为 λ .

$$\begin{aligned} E(X) = D(X) = \lambda \quad E\bar{X} &= E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n} E\left(\sum_{i=1}^n X_i\right) \\ &= \frac{1}{n} \sum_{i=1}^n E(X_i) \\ &= \frac{1}{n} \cdot n\lambda = \lambda \end{aligned}$$

$$\begin{aligned} D(\bar{X}) &= D\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} D\left(\sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \cdot n\lambda = \frac{\lambda}{n} \end{aligned}$$

定理: 设总体 X 的均值为 $E(X) = \mu$.

方差为 $D(X) = \sigma^2$. 样本 $(X_1, X_2, \dots, X_n)$

来自总体 X . 则

$$(1) E(\bar{X}) = \mu \quad (2) D(\bar{X}) = \frac{1}{n} \sigma^2$$

$$(3) E(S^2) = \sigma^2$$

$$\begin{aligned} E(\bar{X}) &= E\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right] = \frac{1}{n}(EX_1 + EX_2 + \dots + EX_n) \\ &= \frac{1}{n} \cdot n\mu = \mu \end{aligned}$$

$$D(\bar{X}) = D\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} D\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$$

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n [(X_i - \mu) - (\bar{X} - \mu)]^2$$

$$= \sum_{i=1}^n [(X_i - \mu)^2 - 2(X_i - \mu)(\bar{X} - \mu) + (\bar{X} - \mu)^2]$$

$$= \sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X} - \mu) \sum_{i=1}^n (X_i - \mu) + \sum_{i=1}^n (\bar{X} - \mu)^2$$

$$= \sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X} - \mu)(n\bar{X} - n\mu) + n(\bar{X} - \mu)^2$$

$$= \sum_{i=1}^n (X_i - \mu)^2 - 2n(\bar{X} - \mu)^2 + n(\bar{X} - \mu)^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

$$\text{即有 } \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

$$E(S^2) = E\left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right]$$

$$= \frac{1}{n-1} E\left[\sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2\right]$$

$$= \frac{1}{n-1} \{E\left[\sum_{i=1}^n (X_i - \mu)^2\right] - nE(\bar{X} - \mu)^2\}$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^n E(X_i - \mu)^2 - nD\bar{X}\right]$$

$$= \frac{1}{n-1} \left(\sum_{i=1}^n D X_i - nD\bar{X}\right)$$

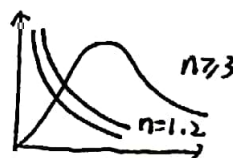
$$= \frac{1}{n-1} (n\sigma^2 - n \cdot \frac{1}{n} \sigma^2)$$

$$= \sigma^2$$

6.3 抽样分布

1. 正态分布

2. χ^2 分布. $\chi^2(n)$ 自由度.



1) $\chi^2(2)$ 是 $\lambda = \frac{1}{2}$ 的指数分布.

2) $n \geq 3$. 单峰曲线. $(n-2)$ 时取最大值关于 $n-2$ 不对称, n 个峰右移.

n 越大, 图形越对称, n 很大的时候可用正态分布近似.

定理: $X_1, \dots, X_n$ 相互独立, 均服从标准正态分布 $N(0, 1)$

$$\chi^2 = \sum_{i=1}^n X_i^2 \sim \chi^2(n)$$

$$E(\chi^2) = n \quad D(\chi^2) = 2n$$

由中心极限定理 $X \sim \chi^2(n)$, n 充分大.

$$\frac{X-n}{\sqrt{2n}} \xrightarrow{\text{近似}} N(0,1)$$

定理: $X \sim \chi^2(n)$, $Y \sim \chi^2(m)$, X, Y 独立.

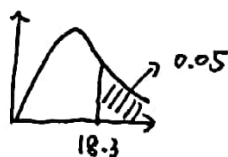
$$X+Y \sim \chi^2(m+n)$$

推论: $X_i \sim \chi^2(m_i)$ 独立, $\sum_{i=1}^n X_i \sim \chi^2(\sum_{i=1}^n m_i)$

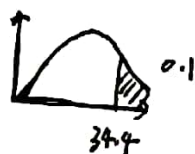
上 α 分位数: $P(\chi^2 > \chi^2_{\alpha}(n)) = \alpha$.
点 面积



$$\chi^2_{0.05}(10) = 18.3$$



$$\chi^2_{0.1}(25) = 34.4$$



例: (1) $X \sim \chi^2(10)$

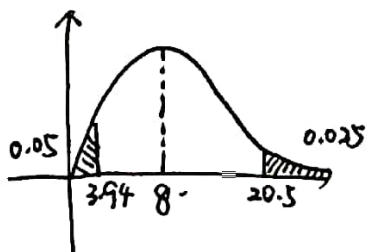
$$P(X > a) = 0.025 \quad P(X < b) = 0.05$$

求 a 和 b .

$n=10$, $\alpha=0.025$ 查表得 $\chi^2_{0.025}(10) = 20.5 = a$.

$$P(X < b) = 1 - P(X > b) \therefore P(X > b) = 0.95$$

$$b = \chi^2_{0.95}(10) = 3.94$$



$$(2) X_1, \dots, X_6 \sim N(0, 2^2)$$

$$\downarrow \downarrow$$

$$u \quad \sigma^2$$

$$\text{求 } P\left(\sum_{i=1}^6 X_i^2 > 6.54\right)$$

(定理: $X_i \sim N(0,1)$) $\sum_{i=1}^n X_i^2 \sim \chi^2(n)$.

$$X_i \sim N(0, 2^2) \quad \frac{X_i - 0}{2} \sim N(0,1)$$

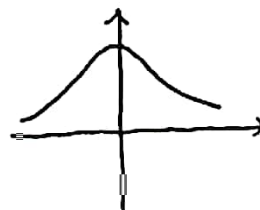
$$\sum_{i=1}^6 \left(\frac{X_i}{2}\right)^2 \sim \chi^2(6)$$

$$P\left(\sum_{i=1}^6 \frac{X_i^2}{2} > \frac{6.54}{2}\right)$$

$$P\left(\frac{\sum_{i=1}^6 X_i^2}{4} > \frac{6.54}{4}\right) = P(\chi^2(6) > 1.635) = 0.95$$

1

t分布 $X \sim t(n)$

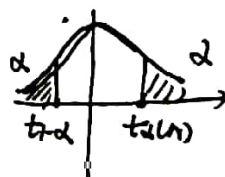


$n \geq 30$ 与正态分布区别很小.

定理: $X \sim N(0,1)$ $Y \sim \chi^2(n)$

$$X, Y \text{ 独立. } \frac{X}{\sqrt{Y/n}} \sim t(n)$$

$$P(T > t_{\alpha}(n)) = \alpha$$



对称性

$$t_{1-\alpha}(n) = -t_{\alpha}(n)$$

例 $X \sim N(2,1)$ $Y_1, Y_2, Y_3, Y_4 \sim N(0, 4)$

$$\text{独立. 令 } T = \frac{4(X-2)}{\sqrt{\sum_{i=1}^4 Y_i^2}}, \text{ 分布?}$$

$$\text{求 } P(T > t_0) = 0.01$$

IV

$$\frac{X-2}{1} \sim N(0,1)$$

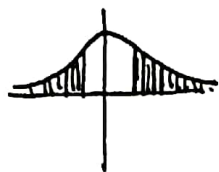
$$\frac{Y_i - 0}{2} \sim N(0,1)$$

$$\sum_{i=1}^4 \left(\frac{Y_i}{2}\right)^2 \sim \chi^2(4)$$

由定理, $\frac{\frac{X-2}{1}}{\sqrt{\frac{\sum_{i=1}^4 (Y_i/2)^2}{4}}} = \frac{4(X-2)}{\sqrt{\sum_{i=1}^4 Y_i^2}} \sim t(4)$

$$P(|T| > t_0) = 0.01$$

$$T > t_0, T < -t_0$$



$$P(T > t_0) = 0.005 \quad n=4 \quad \alpha = 0.005$$

查表得 $t_0 = 4.604$

F分布, $F(n_1, n_2)$

定理 6.5: $X \sim \chi^2(n_1) \quad Y \sim \chi^2(n_2)$

X, Y 独立, $\frac{X/n_1}{Y/n_2} \sim F(n_1, n_2)$

$$\frac{Y/n_2}{X/n_1} \sim F(n_2, n_1)$$

即若 $F \sim F(n_1, n_2) \quad \frac{1}{F} \sim F(n_2, n_1)$

例: $X_1, \dots, X_6 \sim N(0, \sigma^2)$

$$\frac{2(X_1^2 + X_2^2)}{X_3^2 + X_4^2 + X_5^2 + X_6^2}$$

$$\frac{X_i}{\sigma} \sim N(0,1) \quad \frac{X_1^2}{\sigma^2} + \frac{X_2^2}{\sigma^2} \sim \chi^2(2)$$

$$\frac{X_3^2}{\sigma^2} + \frac{X_4^2}{\sigma^2} + \frac{X_5^2}{\sigma^2} + \frac{X_6^2}{\sigma^2} \sim \chi^2(4) \quad \text{变量数}$$

$$\frac{(\frac{X_1^2}{\sigma^2} + \frac{X_2^2}{\sigma^2})/2}{(\frac{X_3^2}{\sigma^2} + \frac{X_4^2}{\sigma^2} + \frac{X_5^2}{\sigma^2} + \frac{X_6^2}{\sigma^2})/4} = \frac{2(X_1^2 + X_2^2)}{X_3^2 + X_4^2 + X_5^2 + X_6^2} \sim F(2, 4)$$

$$P(F > F_{\alpha}(n_1, n_2)) = \alpha$$



$$F_{1-\alpha}(n_1, n_2) = \frac{1}{F_{\alpha}(n_2, n_1)}$$

证: $F \sim F(n_1, n_2) \quad \frac{1}{F} \sim F(n_2, n_1)$

$$P(F > F_{1-\alpha}(n_1, n_2)) = 1 - \alpha$$

$$= P\left(\frac{1}{F} < \frac{1}{F_{1-\alpha}(n_1, n_2)}\right)$$

$$= 1 - P\left(\frac{1}{F} > \frac{1}{F_{1-\alpha}(n_1, n_2)}\right)$$

$$\therefore P\left(\frac{1}{F} > \frac{1}{F_{1-\alpha}(n_1, n_2)}\right) = \alpha$$

$$\frac{1}{F} \sim F(n_2, n_1) \quad F_{\alpha}(n_2, n_1) = \frac{1}{F_{1-\alpha}(n_1, n_2)}$$

例: $F \sim F(10, 15)$ 求 λ_1, λ_2 使

$$P(F > \lambda_1) = 0.01 \quad P(F \leq \lambda_2) = 0.01$$

解: $\lambda_1 = F_{0.01}(10, 15) = 3.8$

$$P(F \leq \lambda_2) = 1 - P(F > \lambda_2) = 0.01$$

$$\therefore P(F > \lambda_2) =$$

$$P(F \leq \lambda_2) = P\left(\frac{1}{F} \geq \frac{1}{\lambda_2}\right) = 0.01$$

$$\frac{1}{F} \sim F(15, 10) \quad P(F \geq \frac{1}{\lambda_2}) = 0.01$$

$$\frac{1}{\lambda_2} = F_{0.01}(15, 10) = 2.193$$

6.3 正态总体下的抽样分布

回忆: 6.2 $X_i \sim N(0,1)$

$$\sum_{i=1}^n X_i^2 \sim \chi^2(n)$$

$$6.3 \quad X \sim \chi^2(n) \quad Y \sim \chi^2(m)$$

$$X+Y \sim \chi^2(m+n)$$

$$6.4. \quad X \sim N(0,1) \quad Y \sim \chi^2(n)$$

$$\frac{X/\sqrt{n}}{\sqrt{Y/n}} \sim F(n, n)$$

$$\frac{X}{\sqrt{Y/n}} \sim t(n)$$

$$6.5 \quad X \sim \chi^2(n_1) \quad Y \sim \chi^2(n_2)$$

$$\frac{X/n_1}{Y/n_2} \sim F(n_1, n_2)$$

总体是正态分布, 从中抽取样本, 构造统计量, $\rightarrow$ 构造出统计量的分布?

一个正态总体

定理: ~~$X \sim N(\mu, \sigma^2)$~~ $X \sim N(\mu, \sigma^2)$, $\{X_1, X_2, \dots, X_n\}$ 为样本

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$(1) \quad \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \quad \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{\bar{X} - \mu}{\sigma} \sqrt{n} \sim N(0,1)$$

$$E\bar{X} = E(\frac{1}{n} \sum_{i=1}^n X_i) = \frac{1}{n} \sum_{i=1}^n EX_i = \frac{1}{n} n\mu = \mu$$

$$D\bar{X} = D(\frac{1}{n} \sum_{i=1}^n X_i) = \frac{1}{n^2} \sum_{i=1}^n DX_i = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}$$

$$\bar{X} = \frac{1}{n} (X_1 + X_2 + \dots + X_n) \quad \text{正态}$$

$$(2) \quad \frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi^2(n-1)$$

$$(3) \quad \bar{X} \text{ 与 } S^2 \text{ 独立}$$

$\uparrow$ 样本均值

定理: $\downarrow$ 总体期望

$$(1) \quad \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \sim \chi^2(n)$$

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi^2(n)$$

标准正态

$$(2) \quad \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim t(n-1)$$

$$\text{证: } \frac{\bar{X} - \mu}{\sigma} \sqrt{n} \sim N(0,1)$$

$$\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi^2(n-1)$$

$$X \sim N(0,1) \quad Y \sim \chi^2(n)$$

$$\frac{X}{\sqrt{Y/n}} \sim t(n)$$

$$\frac{\frac{\bar{X} - \mu}{\sigma} \sqrt{n}}{\sqrt{\frac{(n-1)S^2}{\sigma^2} / (n-1)}} \sim t(n-1)$$

两个正态总体

$$X \sim N(\mu_1, \sigma_1^2) \quad Y \sim N(\mu_2, \sigma_2^2)$$

$$\{X_1, \dots, X_{n_1}\} \quad \{Y_1, Y_2, \dots, Y_{n_2}\}$$

$$\bar{X}, S_1^2 \quad \bar{Y}, S_2^2$$

$$(1) \quad \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$$

$$\bar{X} \sim N(\mu_1, \frac{\sigma_1^2}{n_1}) \quad \bar{Y} \sim N(\mu_2, \frac{\sigma_2^2}{n_2})$$

$$\bar{X} - \bar{Y} \sim N(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$$

$$(2) \quad \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1)$$

$$\frac{(n_1-1)S_1^2}{\sigma_1^2} \sim \chi^2(n_1-1) \quad \frac{(n_2-1)S_2^2}{\sigma_2^2} \sim \chi^2(n_2-1)$$

$$\frac{\frac{(n_1-1)S_1^2}{\sigma_1^2} / (n_1-1)}{\frac{(n_2-1)S_2^2}{\sigma_2^2} / (n_2-1)} \sim F(n_1-1, n_2-1)$$

VI

$$\sigma_1^2 = \sigma_2^2 = \sigma^2 \text{ 时: } T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1+n_2-2)$$

例: $X \sim N(\mu, \sigma^2)$ $\bar{X}$ 为样本均值

S^2 为样本方差, 样本容量 $n=16$

求 k 使 $P(\bar{X} > \mu + kS) = 0.95$

解: $\frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{4(\bar{X} - \mu)}{S} \sim t(16-1) = t(15)$

$$P(\bar{X} > \mu + kS) = P\left(\frac{4(\bar{X} - \mu)}{S} > \frac{4kS}{S}\right) = 0.95$$

查表得 $4k = -1.753$ $k = -0.438$

例: $X \sim N(\mu, \sigma^2)$ $(x_1, \dots, x_{n+1})$ 为样本

$\bar{X}, S_n^2$ 是 $(x_1, \dots, x_n)$ 的均值和方差.

$\frac{X_{n+1} - \bar{X}}{S_n} \sqrt{\frac{n}{n+1}}$ 的分布

解: $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ $X_{n+1} \sim N(\mu, \sigma^2)$

$$X_{n+1} - \bar{X} \sim N(0, \frac{\sigma^2}{n} + \sigma^2)$$

$$U = \frac{X_{n+1} - \bar{X}}{\sqrt{\frac{\sigma^2}{n} + \sigma^2}} \sim N(0, 1)$$

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1)$$

$$\text{由 } \frac{X}{\sqrt{Y/n}} \sim t(n)$$

$$\frac{\frac{X_{n+1} - \bar{X}}{\sigma \sqrt{\frac{n+1}{n}}}}{\sqrt{\frac{(n-1)S_n^2}{\sigma^2}/(n-1)}} \sim t(n-1)$$

$$\therefore \frac{X_{n+1} - \bar{X}}{S_n} \sqrt{\frac{n}{n+1}} \sim t(n-1)$$